Date: 23rd Sep, 25 BDA Day: Tuesday

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Rg No: FA23-BBD-086

Assignment: 01
Chronic Disease Analysis

① Importing Libraries:

```
import pandas as pd
import numpy as np
import statistics as st

chronic_diseases = pd.read_csv('chronic_disease.csv')
chronic_diseases.head(15)
chronic_diseases.tail(15)
```

a) What are the types of variables in the data set?

By analyzing the last and first rows (15) of the dataset there are a total of 10 variables (unique, categorical and numeric) in the dataset.

Their explanation is given below:

① Categorical Variables:

There are six categorical variables in the dataset and they are:

i. Patient_Number: A unique identifier for each patient like registration number of students in a university.

ii. BP_Abnormality: Indicating patients that do and do not have blood pressure.
0: No Blood Pressure
1: Blood Pressure exists.

iii. Females: Gender variable that contains two categories likely 0: Male and 1: Females.

iv. Smoking: Indicating patients that are non-smoker: 0 and smoker: 1.

v. Chronic_Kidney_Disease:
Status indicating patients that have kidney disease (1) and those who do not have (1).

vi. Adrenal_thyroid_disorders:
Indicating patients with disorder (1) and no disorder (0).

b) Numerical variables:

i. Level_of_Hemoglobin: Indicating patients with various hemoglobin levels ranging from 11.28 ~~11.36~~. and ending at 11.36

ii. Age: Age of every patient from 34 - 41 (first observation to least observation).

iii. Step_Walk_Week:
Indicating patients that walk specific steps per week.

iv. BMI:
Body Mass Index (BMI) level for each patient.

b) Descriptive Analysis
Variables that I chose:
① Females (Categorical Variable)
② BMI (Numerical variable).

chronic_diseases.groupby('Female')['BMI'].describe()

There are total 1008 females and 992 males. Males have slightly average value of BMI (30.32) as compared to females (29.96)

There is little to no difference in mean and median value, indicating that there are little to less outliers in the dataset.

25% of both males and females have BMI value less than 20, and 75% have less than 40 or 25% have BMI value greater than 40.

The spreadness for males (11.9155) is almost identical to the spreadness for females. The difference is ~~neglib~~ negligible. This indicates that the spreadness for both genders are distributed equally around their means.

```
chronic_disease.groupby('Female')['BMI'].skew()
```

The skewness interprets on the assymetry of the variables you choose for analysis. Since, the skewness is negative for both gender (Male and Females), this implies that data is dispersed to the left (left-skewness), but since the values (-0.000230 & -0.033094) it means that data is at the centre.

(Single) Group By Analysis

```
chronic_diseases.groupby('Females').mean(['Patient_Number'
```

Female patients a slightly older (47) than males (45), have lower hemoglobin (10) than males (12), and show a higher prevalence of blood pressure abnormality (52%) than males (46%) and ~~key~~ kidney disease (~~0.51%~~) is slightly more for females (0.51%) than males. BMI, smoking habits and walking activity are nearly same across genders. Adrenal and thyroid disorders a ~~a~~ little more common in males.